Que algebra of type $A^{(1)}_{N-1}$

$V = \mathbb{C}^N$ vector rep

$U_q\widehat{\mathfrak{sl}}_N$ is the unital algebra over $\mathbb{C}$ generated by

$e_i, f_i, K_i^{\pm 1}$, and relations - - -

$i = 0, 1, \dots, N-1$

$\Delta(e_i) = e_i \otimes 1 + K_i \otimes e_i, \quad \Delta(f_i) = f_i \otimes K_i^{-1} + 1 \otimes f_i$

$\Delta(K_i^{\pm 1}) = K_i^{\pm 1} \otimes K_i^{\pm 1}$

$U_q\mathfrak{sl}_n$: $K_0 K_1 \cdots K_{N-1} = 1$

$\mathbb{Z}^N = \bigoplus_{i \in \mathbb{Z}/N\mathbb{Z}} \mathbb{Z}\epsilon_i$ (0) weight lattice

~~$\alpha_i = \epsilon_i - \epsilon_{i+1}$ (simple roots)~~, $\alpha_0 = \epsilon_N - \epsilon_1$

$U_q L\mathfrak{sl}_n$: adjoint generators t_λ, $\lambda \in \mathbb{Z}^N$

$t_\lambda t_\mu = t_{\lambda+\mu}$, $t_0 = 1$, $t_\lambda e_i t_{-\lambda} = q^{(\lambda|\alpha_i)} e_i$

$t_\lambda f_i t_{-\lambda} = q^{-(\lambda|\alpha_i)} f_i$

$t_i = t_{\epsilon_i}$

So that $K_i = t_i t_{i+1}^{-1}$

Hopf-algebra Automorphism of $U_q\hat{\mathfrak{sl}}_N$ ϕ_z $(z \in \mathbb{C}^\times)$

↓ identity on all generators except e_0, f_0

↓ U_q

$\phi_z(e_0) = z e_0$, $\phi_z(f_0) = z^{-1} f_0$

Exterior Powers and Clifford Algebra

Let $V = \mathbb{C}^N$ with basis $v_1, \cdots, v_N$.

C_N be Clifford algebra of $V \oplus V^*$ with quadratic form

$$Q(v \oplus \alpha) = \alpha(v)$$

↓ unital algebra generated by ψ_i^*, ψ_i (images of basis vectors), $i \in \{1, \cdots, N\}$

s.t. $\psi_i \psi_j = -\psi_j \psi_i$, $\psi_i^* \psi_j^* = -\psi_j^* \psi_i^*$, $\psi_i \psi_j^* + \psi_j^* \psi_i = \delta_{ij} \mathbb{1}$. ——(1)

* $\wedge V$ is C_N-module

↳ $\psi_i^* \cdot v = v_i \wedge v$, $\psi_i \to$ contraction

↳ generated by $|0\rangle$

* $n_i = \psi_i^* \psi_i$ have common eigenbasis $v_{i_1} \wedge \cdots \wedge v_{i_k}$ $(i_1 < \cdots < i_k)$

with eigenvalue 1 if $i = i_j$ for some j

0 otherwise

* $t_i^{\pm 1} := q^{\pm n_i} = q^{\pm 1} \psi_i^* \psi_i + \psi_i \psi_i^*$

$\to t_i t_i^{-1} = 1$, $t_i t_j = t_j t_i$, $t_i \psi_j^* t_i^{-1} = q^{\delta_{ij}} \psi_j^*$, $t_i \psi_j t_i^{-1} = q^{-\delta_{ij}} \psi_j$.

$C_N(q) = \langle t_i, \psi_i, \psi_i^*, i = 1, \cdots, N \rangle / (1), (2)$. $\forall i, j$. ——(2)

$C_N(q)$ is $\mathbb{Z}_2$-graded : ψ_i, ψ_i^* odd, t_i even

ΛV – $\mathbb{Z}_2$-graded module : $|0\rangle$ is even

Proposition (Hayashi)

$\exists$ a homomorphism $U_q\widehat{\mathfrak{gl}}_N \longrightarrow C_N(q)$ s.t.

$$e_i \mapsto \psi_i^* \psi_{i+1}, \quad f_i \mapsto \psi_{i+1}^* \psi_i, \quad t_i^{\pm 1} \mapsto t_i^{\pm 1}, \quad i \in \mathbb{Z}_N$$

$\Lambda^\circ V(z)$: $e_0 \mapsto z\psi_N^* \psi_1$, $f_0 \mapsto z^{-1}\psi_1^* \psi_N$.

$\Lambda V \otimes \Lambda V \longrightarrow$ tensor prod. of $\mathbb{Z}_2$-graded modules.

tensor prod. of linear maps : $(f\otimes g)(v\otimes w) = (-1)^{|g||v|} fv \otimes gw$

$$E = \sum_{i=1}^{N} (\psi_i^* \otimes \psi_i) \prod_{j>i} K_j, \quad F = -\sum_{i=1}^{N} \prod_{j<i} K_j^{-1} (\psi_i \otimes \psi_i^*), \quad K = \prod_{i=1}^{N} K_i,$$

$$K_i = t_i \otimes t_i^{-1}, \quad i = 1, \ldots, N.$$

Lemma

E, F, K commute with $U_q\widehat{\mathfrak{gl}}_N$-action on $\Lambda V \otimes \Lambda V$ and obey relations

$$[E,F] = \frac{K-K^{-1}}{q-q^{-1}}, \quad KEK^{-1} = q^2 E, \quad KFK^{-1} = q^{-2} F.$$

$$T_1 = \prod_{i=1}^{N} t_i \otimes 1, \quad T_2 = \prod_{i=1}^{N} 1 \otimes t_i.$$

Theorem

The braiding matrix

$$\overset{\vee}{R}_{k,k'}(z_1/z_2) : \Lambda^k V(z_1) \otimes \Lambda^{k'} V(z_2) \longrightarrow \Lambda^{k'} V(z_2) \otimes \Lambda^k V(z_1),$$

normalized so that $\overset{\vee}{R}_{k,k'}\left(\frac{z_1}{z_2}\right) v^k \otimes v^{k'} = (-1)^{kk'} v^{k'} \otimes v_k$

is given by $\overset{\vee}{R}_{k,k'}\left(\frac{z_1}{z_2}\right) = (-1)^{\min(k,k')} A_{k-k'}\left(\frac{z_1}{z_2}\right)$

where $A_{k-k'}\left(\frac{z_1}{z_2}\right)$ is ~~given by~~ defined by

$$A_{\overset{\vee}{R}_{k,k'}}(z) = \sum_{j=0}^{n} (-q)^j \frac{1-q^{|m|}z}{1-q^{2j+|m|}z} \begin{cases} E^{(j)} F^{(j+m)} & \text{if } m \geq 0 \\ E^{(j-m)} F^{(j)} & \text{if } m \leq 0 \end{cases}$$

$\underline{m = k-k'}$

with $E^{(\ell)} = \frac{E^\ell}{[\ell]_q!}$, $F^{(\ell)} = \frac{F^\ell}{[\ell]_q!}$

and $n = \min(k, k', N-k, N-k')$

Proof :—

LHS $= \Lambda(V \otimes \mathbb{C}^2) = \Lambda(V \oplus V) = \Lambda V \otimes \Lambda V = \bigoplus^{N} \Lambda^k V \otimes \Lambda^{k'} V$

$k,k'=0$

RHS Comparing the (k,k') $U_q\mathfrak{sl}_2$-weight space of weight (k,k') for $U_q\mathfrak{sl}_2$ action

we have :—

$$\Lambda^k V \otimes \Lambda^{k'} V \cong \bigoplus_{s=0}^{n} V^N_{(k+s,\, k'-s)^t} \qquad (k \geq k')$$

$n = \min(k, k', N-k, N-k')$

Similar reasoning when $k < k'$

Let $v^K = \psi_K^* \psi_{K-1}^* \cdots \psi_1^* |0\rangle$

$s=0$: $v_0^{(K,K')} = v^K \otimes v^{K'}$ is singular.

In general the singular vector $v_s^{(K,K')}$ is

Proportional to $F^{(s)} v_0^{(K+s, K'-s)}$ if $K \geq K'$

$F^{(K'-K+s)} v_0^{(K'+s, K-s)}$ if $K \leq K'$

Normalize so that

$$v_s^{(K,K')} = \psi_K^* \cdots \psi_1^* |0\rangle \otimes \psi_{K+s}^* \cdots \psi_{K+1}^* \psi_{K'-s}^* \cdots \psi_1^* |0\rangle + \dots$$

Let $Q_s : \Lambda^K V \otimes \Lambda^{K'} V \longrightarrow \Lambda^{K'} V \otimes \Lambda^K V$

be unique $U_q \mathfrak{gl}_N$-linear map so that

$$v_s^{(K,K')} \longmapsto v_{s'}^{(K',K)}, \quad s' = K - K' + s$$

same weight $(K+s, K'-s) = (K'+s, K-s)$

and other irred. comp. to 0.

$$R_{K,K'}(z) = \sum_{s=0}^{n} (-1)^{KK' + (K-K')s} \prod_{j=1}^{s} \frac{z - q^{|K-K'|+2j}}{1 - z q^{|K-K'|+2j}} Q_s.$$

$\longrightarrow$ Compare normalizations of singular vectors.

$$\vartheta_s^{(k,k')} = (-1)^{sk}\, q^{ss'}\, F^{(s)}\, \vartheta_0^{(k+s,k'-s)}$$

inverse of

$$\prod_{i=1}^{s}\left[(-1)(-1)^{s-i}(-1)^{k+s+1-\ell}\right] \qquad \left[(q^{-1})^{k}\, q^{k'-s}\right]^{s}$$

Divide $F^{(s)}\vartheta_0^{(k+s,k'-s)}$ by this to get $\vartheta_s^{(k,k')}$

Similarly, $\vartheta_{s'}^{(k',k)} = (-1)^{s'k'} q^{ss'} F^{(s')} \vartheta_0^{(k+s,k'-s)}$

$k' \le k \Rightarrow \vartheta_{s'}^{(k',k)} \propto F^{(k-k'+s)} \vartheta_0^{k+s,k'-s}$

$= F^{(s')} \vartheta_0^{k'+s',\,k-s'}$

Thus, for $k \ge k'$

$$\check{R}_{kk'}(z)\, F^{(s)} \vartheta_0^{(k+s,k'-s)} = (-1)^{k'} \prod_{j=1}^{s} \frac{z - q^{|k-k'|+2j}}{1 - z q^{|k-k'|+2j}}\, F^{(s')} \vartheta_0^{(k+s,k'-s)}$$

<u>Lemma</u>

$$A_m(z)\, \vartheta_m^{\ell} = \prod_{j=0}^{\frac{\ell-|m|}{2}-1} \frac{z - q^{\ell-2j}}{1 - z q^{\ell-2j}}\, \vartheta_{-m}^{\ell}$$

$\ell = k-k'+2s$, $m = k-k'$

$\ell - 2j = k-k'+2(s-j)$

$j \mapsto s-j$

Similar reasoning when $k \le k'$

Proof of Lemma

$$C = EF + \frac{q^{-1}K + qK^{-1}}{(q-q^{-1})^2} = FE + \frac{qK + q^{-1}K^{-1}}{(q-q^{-1})^2}$$

$$C \cdot L_\ell := C_\ell = \frac{q^{\ell+1} + q^{-\ell-1}}{(q-q^{-1})^2}$$

$$E^j F^j = \prod_{i=0}^{j-1} \left(C - \frac{q^{2i+1}K^{-1} + q^{-2i-1}K}{(q-q^{-1})^2} \right)$$

$$E^j F^j \Big|_{L_\ell[-\ell+2K]} = \prod_{i=0}^{j-1} [\ell+1-K+i][K-i]$$

$$\Rightarrow A_m(z) = \sum_{j=0}^{K} (-q)^j \frac{1-q^m z}{1-q^{2j+m} z} \frac{1}{[j]_q! \, [j+m]_q!} \prod_{i=0}^{j-1} [\ell+1-K-i][K-i] F^m$$

$$\Rightarrow A_m(z) v_m^\ell = \sum_{j=0}^{K} (-q)^j \frac{1-q^m z}{1-q^{2j+m} z} \frac{[\ell-K]_q!}{[j]_q! \, [j+m]! \, [K]!} \prod_{i=0}^{j-1} [\ell+1-K-i][K-i] v_m^\ell \quad (11)$$

$$= \sum_{j=0}^{K} (-q)^j \frac{1-q^{\ell-2K} z}{1-q^{2j+\ell-2K} z} \prod_{i=0}^{K-1} \frac{[\ell-K+j-i]}{[j][K-j]} v_{-m}^\ell$$

↙ rational fn. with simple poles at $z = q^{-\ell+2K-2}, q^{-\ell+2K-4}, \ldots, q^{-\ell}$ (bounded at infinity)

Comparing residues

U be f.d. $U_q\mathfrak{gl}_M$-module of type 1.

$$z = (z_1, z_2, \dots, z_M)$$

$$\mu = (\mu_1, \mu_2, \dots, \mu_M) \in \mathbb{Z}^M$$

$$A_{i,\mu}(z) = \sum_{j=0}^{\infty} (-q)^j \frac{z_{i+1} - q^{|\mu_i - \mu_{i+1}|} z_i}{z_{i+1} - q^{|\mu_i - \mu_{i+1}|+2} z_i} E_i^{(j+(\mu_{i+1}-\mu_i)_+)} F_i^{(j+(\mu_{i+1}-\mu_i)_+)}$$

$(x)_+ = \max\{x, 0\}$.

↓ rational fn. in M variables with values in $U_q\mathfrak{sl}_M \subset U_q\mathfrak{gl}_M$

$$A_{i,\mu}(z) = \jmath_i\left(A_{\mu_i,\mu_{i+1}}\left(\frac{z_i}{z_{i+1}}\right)\right)$$

$\jmath_i : U_q\mathfrak{sl}_2 \hookrightarrow U_q\mathfrak{sl}_M$

$E, F, K^{\pm 1} \longmapsto E_i, F_i, K_i^{\pm 1}$

$$A_{i,\mu}(z) : U[\mu] \longrightarrow U[s_i\mu]$$

Theorem

1) $A_{i, s_j\mu}(s_j z) A_{j,\mu}(z) = A_{j, s_i\mu}(s_i z) A_{i,\mu}(z) \in \mathrm{Hom}(U[\mu], U[s_i s_j \mu])$

$1 \le i, j \le M-1$ and $|i-j| \ge 2$

2) $A_{i, s_i\mu}(s_i z) A_{i,\mu}(z) = \mathrm{id} \in \mathrm{End}(U[\mu])$

3) $A_{i, s_{i+1}s_i\mu}(s_{i+1}s_i z) A_{i+1, s_i\mu}(s_i z) A_{i,\mu}(z)$

$= A_{i+1, s_i s_{i+1}\mu}(s_i s_{i+1} z) A_{i, s_{i+1}\mu}(s_{i+1} z) A_{i+1,\mu}(z)$

$\in \mathrm{Hom}(U[\mu], U[s_i s_{i+1} s_i \mu])$

Let $U = \bigoplus_{\mu} U[\mu]$

$$A_{i,U}(z)\Big|_{U[\mu]} = A_{i,\mu} : U[\mu] \to U[s_i\mu]$$

$A_{i,U}(z)$: End(U)-valued rational fn } $U(z)$

<u>Corollary</u>

$$(s_i f)(z) = A_{i,U}(s_i z)\, f(s_i z) \quad , \; i = 1, \dots, M-1$$

defines a rep of S_M on $U(z)$.